

Report Period	Sep 14, 2026 12:02 - Sep 14, 2026 12:08
Generated At	2026-09-25 14:18:26
Classification	ML: Isolation Forest + Random Forest

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		16467	
Total Threats Mitigated	20	Total Packets Analyzed		3516546809	
High Priority	4	Malicious Packets		3512904731	
Low Priority	16	True Negatives Passed		11563	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		0	
ML Metrics	Value	SYN Flood		13	
False Positives	10	UDP Flood		3	
FP Rate	0.08%	Uncertain		4	
IF Precision	99.82%				
IF Recall	91.20%	Action		Count	
IF F1-Score	95.31%	Quarantined		5	
IF Accuracy	97.08%	Time Ban		4	
		Blackhole		0	
		Released		2	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.82%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	91.20%	Share of actual attacks successfully flagged
F1-Score	95.31%	Balanced measure combining Precision and Recall
Accuracy	97.08%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.08%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	8.80%	Actual attacks that went undetected
True Positive Rate (TPR)	91.20%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.92%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4882	FN 471
Actual: Normal	FP 9	TN 11092

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	4882	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	0

2c. Hybrid Cascaded

Metric	Value	Description
Precision	99.82%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	91.20%	Share of actual attacks successfully flagged
F1-Score	95.31%	Balanced measure combining Precision and Recall
Accuracy	97.08%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.08%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	8.80%	Actual attacks that went undetected
True Positive Rate (TPR)	91.20%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.92%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4882	FN 471
Actual: Normal	FP 9	TN 11092

2d. Response Latency

Metric	Value	Description
Detection Time	314.0 ms	Average time taken to detect an attack
Mitigation Response Time	0.6 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	0.90%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	4.76%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	7.95%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.6	4	4	2026-09-14 12:03:13	2026-09-14 12:06:35
10.0.0.16	3	3	2026-09-14 12:06:02	2026-09-14 12:07:24
10.0.0.17	2	2	2026-09-14 12:06:07	2026-09-14 12:07:18
10.0.0.18	2	2	2026-09-14 12:06:11	2026-09-14 12:07:22
10.0.0.19	2	2	2026-09-14 12:06:03	2026-09-14 12:07:14

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-14 12:03:13	10.0.0.6	DDoS	UDP Flood	92.0%	High	Time Ban (1m)
2026-09-14 12:04:14	10.0.0.6	DDoS	UDP Flood	92.0%	High	Released
2026-09-14 12:04:24	10.0.0.6	DDoS	UDP Flood	92.0%	High	Quarantined
2026-09-14 12:06:02	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-14 12:06:03	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-14 12:06:07	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-14 12:06:11	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-14 12:06:13	10.0.0.16	DDoS	SYN Flood	92.6%	Critical	Time Ban (1m)
2026-09-14 12:06:14	10.0.0.19	DDoS	SYN Flood	92.3%	Critical	Time Ban (1m)
2026-09-14 12:06:14	10.0.0.16	DDoS	SYN Flood	92.6%	Critical	Time Ban
2026-09-14 12:06:14	10.0.0.19	DDoS	SYN Flood	92.3%	Critical	Time Ban
2026-09-14 12:06:18	10.0.0.17	DDoS	SYN Flood	93.1%	Critical	Time Ban (1m)
2026-09-14 12:06:18	10.0.0.17	DDoS	SYN Flood	93.4%	Critical	Time Ban
2026-09-14 12:06:22	10.0.0.18	DDoS	SYN Flood	93.1%	Critical	Time Ban (1m)
2026-09-14 12:06:23	10.0.0.18	DDoS	SYN Flood	93.1%	Critical	Time Ban
2026-09-14 12:07:13	10.0.0.16	DDoS	SYN Flood	93.0%	Critical	Released
2026-09-14 12:07:14	10.0.0.19	DDoS	SYN Flood	92.3%	Critical	Time Ban (2m)
2026-09-14 12:07:18	10.0.0.17	DDoS	SYN Flood	93.4%	Critical	Time Ban (2m)
2026-09-14 12:07:22	10.0.0.18	DDoS	SYN Flood	93.1%	High	Time Ban (2m)
2026-09-14 12:07:24	10.0.0.16	DDoS	SYN Flood	93.0%	Critical	Time Ban (2m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.16	SYN Flood	0.6365	93.0%	Critical	Phase 2	3	11s	2026-09-14 12:07:24	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6057	93.1%	High	Phase 2	2	1m 11s	2026-09-14 12:07:22	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6182	93.4%	Critical	Phase 2	2	1m 11s	2026-09-14 12:07:18	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6396	92.3%	Critical	Phase 2	2	1m 11s	2026-09-14 12:07:14	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6295	93.0%	Critical	Phase 2	1	1m 11s	2026-09-14 12:07:13	Ban Expired
10.0.0.6	UDP Flood	0.5690	92.0%	High	Phase 1	4	10s	2026-09-14 12:06:35	Attack Stopped

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.18	SYN Flood	0.6417	93.1%	Critical	Phase 2	1	11s	2026-09-14 12:06:22	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6420	93.1%	Critical	Phase 2	1	11s	2026-09-14 12:06:18	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6490	92.3%	Critical	Phase 2	1	11s	2026-09-14 12:06:14	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6392	92.6%	Critical	Phase 2	1	11s	2026-09-14 12:06:13	Attack persisted - escalating to time ban
10.0.0.6	UDP Flood	0.6098	92.0%	High	Phase 1	3	10s	2026-09-14 12:05:22	Attack Stopped
10.0.0.6	UDP Flood	0.6317	92.0%	High	Phase 1	2	11s	2026-09-14 12:04:35	Attack Stopped
10.0.0.6	UDP Flood	0.6045	92.0%	High	Phase 2	1	1m 1s	2026-09-14 12:04:14	Ban Expired

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE